

A decorative graphic on the right side of the slide, consisting of a grid of squares. Each square is divided diagonally from the top-left to the bottom-right. The top-left half of each square is a solid green color, while the bottom-right half is a lighter shade of green. The squares are arranged in a staggered, overlapping pattern that creates a sense of depth and movement, resembling a stylized staircase or a modern architectural element.

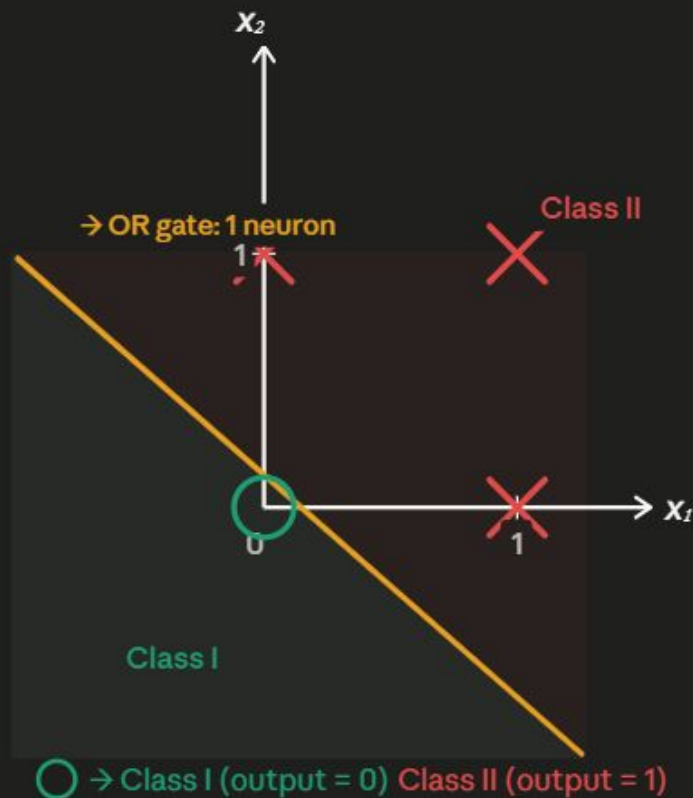
Neural Network Artificial Intelligence

OR Gate Classification Problem

x_1	x_2	Output	$net = x_1 + x_2$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	1	2

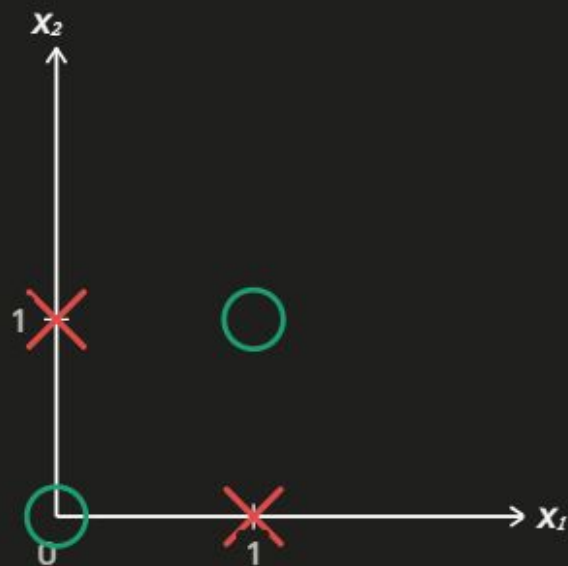
0 = Class I

1 = Class II



XOR Classification Problem

x_1	x_2	Output	$net = x_1 + x_2$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	2

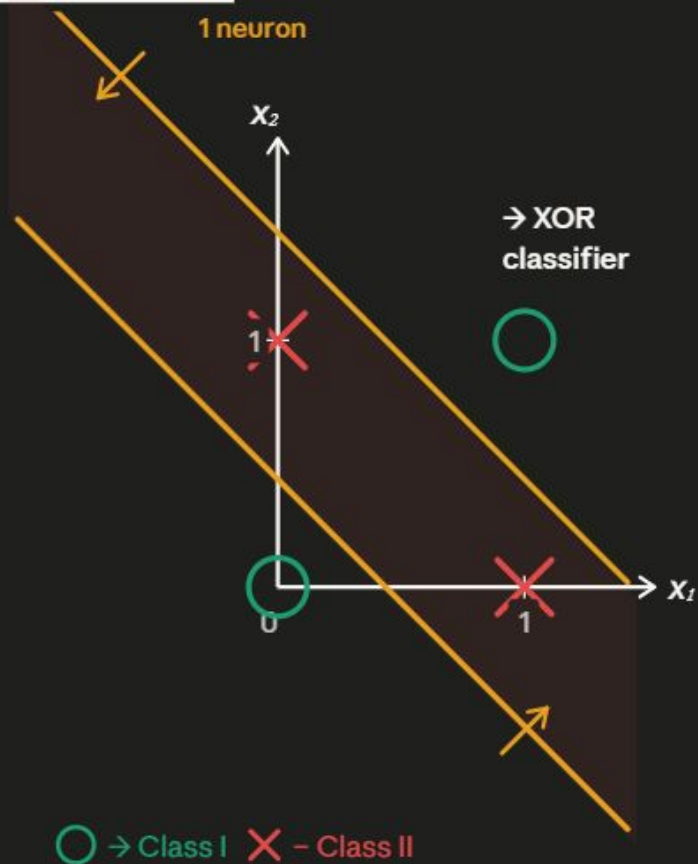


$\bigcirc \rightarrow$ Class I

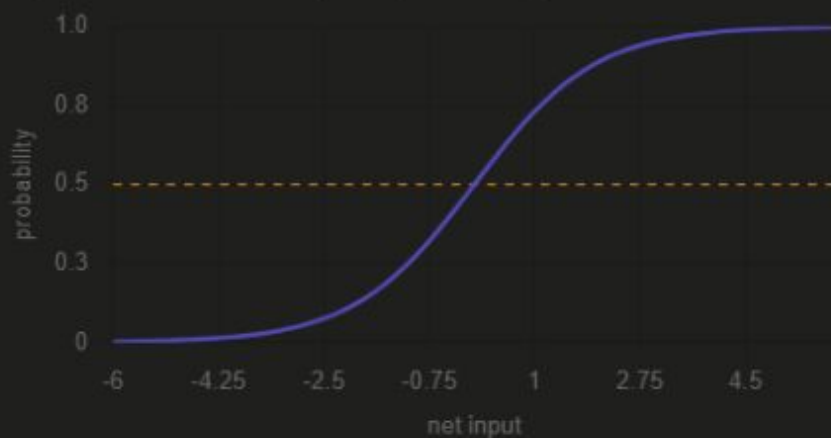
$\times$ - Class II

XOR Classification Problem

x_1	x_2	Output	$net = x_1 + x_2$
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	2



Sigmoid curve — net input $\rightarrow$ probability



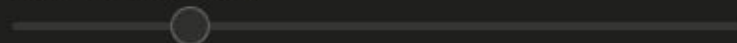
OR gate — 4 data points with probabilities

x_1	x_2	net	P(class II)	Output
0	0	-1.0	0.27	0 ✓
0	1	2.0	0.88	1 ✓
1	0	2.0	0.88	1 ✓
1	1	5.0	0.99	1 ✓

Decision boundary ($x_1 + x_2 = 0.5$)

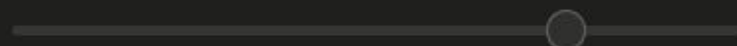


Adjust weight (w)



Higher weight $\rightarrow$ steeper sigmoid $\rightarrow$ sharper boundary

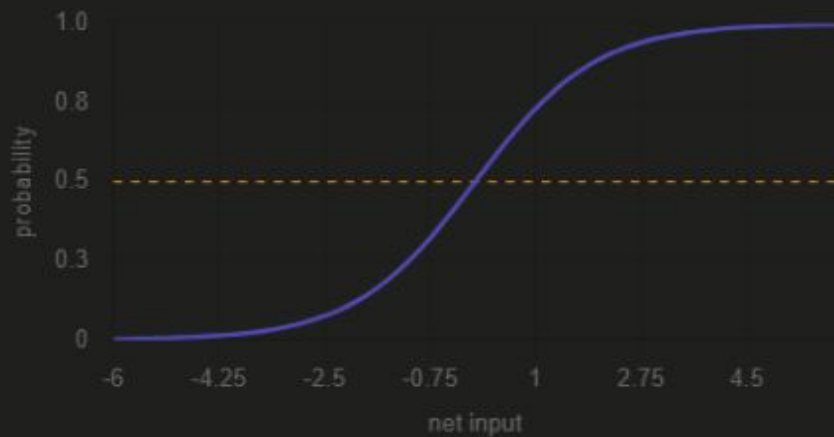
Bias



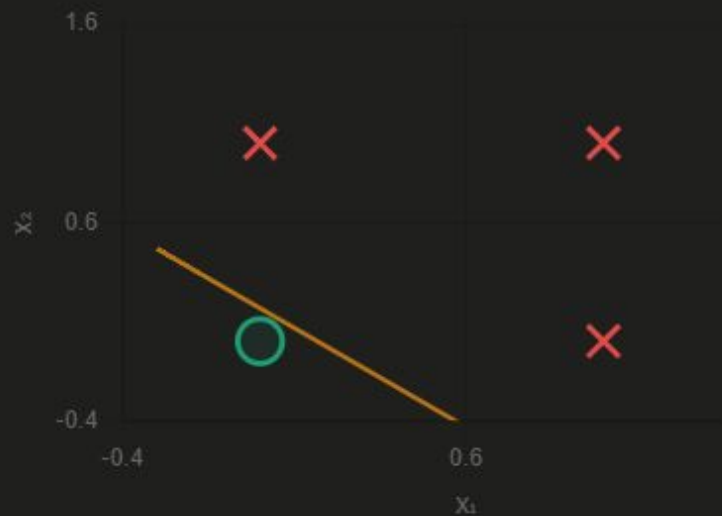
Bias shifts the boundary left/right

○ Class I (OR=0) × Class II (OR=1) — Boundary

Sigmoid curve — net input $\rightarrow$ probability



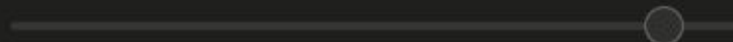
Decision boundary ($x_1 + x_2 = 0.5$)



OR gate — 4 data points with probabilities

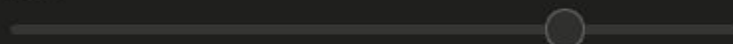
x_1	x_2	net	P(class II)	Output
0	0	-1.0	0.27	0 ✓
0	1	8.0	1.00	1 ✓
1	0	8.0	1.00	1 ✓
1	1	17.0	1.00	1 ✓

Adjust weight (w)



Higher weight $\rightarrow$ steeper sigmoid $\rightarrow$ sharper boundary

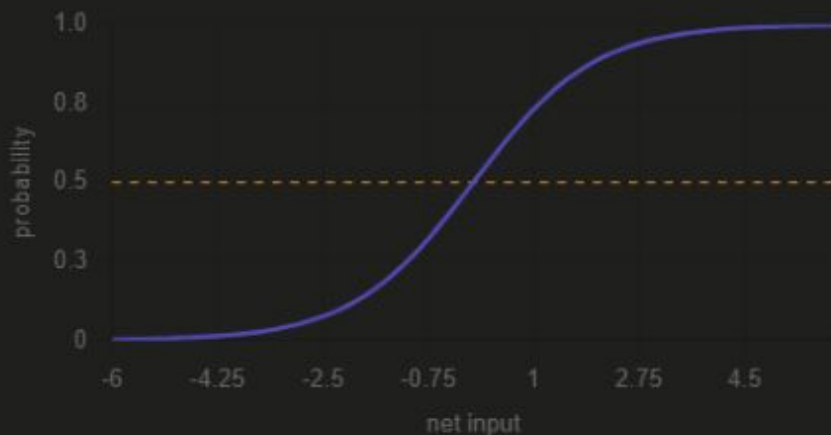
Bias



Bias shifts the boundary left/right

○ Class I (OR=0) × Class II (OR=1) — Boundary

Sigmoid curve — net input $\rightarrow$ probability



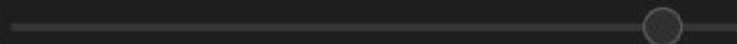
Decision boundary ($x_1 + x_2 = 0.5$)



OR gate — 4 data points with probabilities

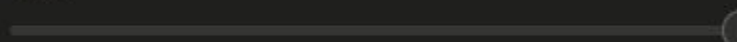
x_1	x_2	net	P(class II)	Output
0	0	0.0	0.50	1 ✗
0	1	9.0	1.00	1 ✓
1	0	9.0	1.00	1 ✓
1	1	18.0	1.00	1 ✓

Adjust weight (w)



Higher weight $\rightarrow$ steeper sigmoid $\rightarrow$ sharper boundary

Bias



Bias shifts the boundary left/right

○ Class I (OR=0) ✗ Class II (OR=1) — Boundary

Model: $w = 2$, bias = -5 | net = $2x - 5$ | $\hat{y} = \sigma(\text{net}) = 1/(1+e^{-\text{net}})$

Step 1

Compute net input

x	$2x - 5$	net
1	$2(1) - 5$	-3
2	$2(2) - 5$	-1
3	$2(3) - 5$	+1
4	$2(4) - 5$	+3

Step 2

Apply sigmoid $\rightarrow \hat{y}$

net	calculation	$\hat{y}$
-3	$1/(1+e^3)$	0.05
-1	$1/(1+e^1)$	0.27
+1	$1/(1+e^{-1})$	0.73
+3	$1/(1+e^{-3})$	0.95

Step 3

Threshold at 0.5 $\rightarrow$ predict

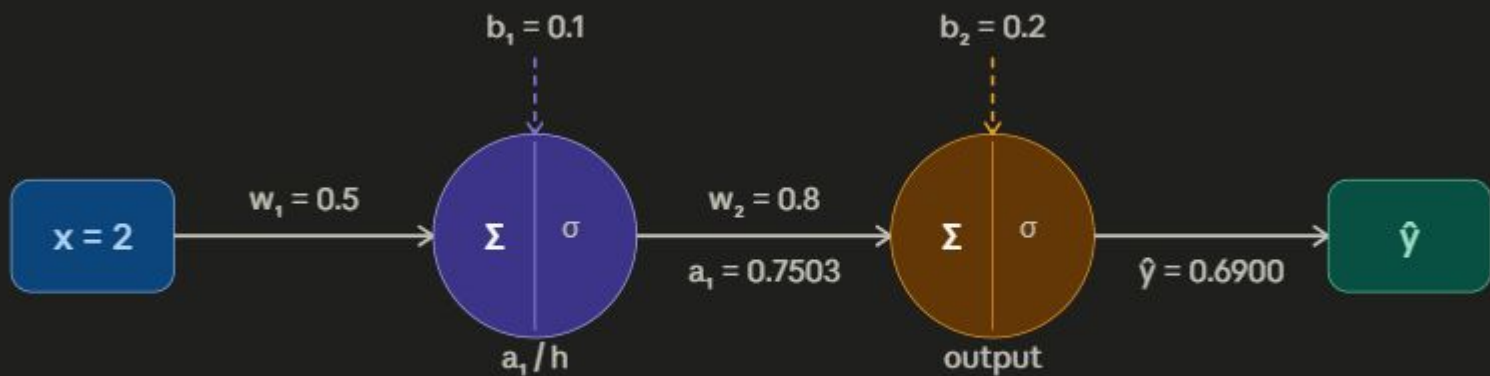
x	$\hat{y}$	Actual	OK?
1	$0.05 \rightarrow 0$	0 Fail	✓
2	$0.27 \rightarrow 0$	0 Fail	✓
3	$0.73 \rightarrow 1$	1 Pass	✓
4	$0.95 \rightarrow 1$	1 Pass	✓

Decision boundary: $2x - 5 = 0 \rightarrow x = 2.5$ hours

Accuracy: 4 / 4 correct (100%)

Given Values

Parameter	Value
Input	$x = 2$
True label	$y = 1$
w_1	0.5
b_1	0.1
w_2	0.8
b_2	0.2
Learning rate η	0.5



$$\begin{aligned} net_1 &= w_1 x + b_1 = 1.1 \rightarrow a_1 = \sigma(1.1) = 0.7503 \\ net_2 &= w_2 \cdot a_1 + b_2 = 0.8002 \rightarrow \hat{y} = \sigma(0.8002) = 0.6900 \end{aligned}$$

FORWARD PASS

Step 1: Layer 1 Pre-activation

$$z_1 = w_1 \cdot x + b_1 = 0.5 \times 2 + 0.1 = 1.0 + 0.1 = \boxed{1.1}$$

Step 2: Layer 1 Activation (Sigmoid)

$$a_1 = \sigma(z_1) = \frac{1}{1 + e^{-1.1}} = \frac{1}{1 + 0.3329} = \frac{1}{1.3329} = \boxed{0.7503}$$

Step 3: Layer 2 Pre-activation

$$z_2 = w_2 \cdot a_1 + b_2 = 0.8 \times 0.7503 + 0.2 = 0.6002 + 0.2 = \boxed{0.8002}$$

Step 4: Output (Sigmoid)

$$\hat{y} = \sigma(z_2) = \frac{1}{1 + e^{-0.8002}} = \frac{1}{1 + 0.4493} = \frac{1}{1.4493} = \boxed{0.6900}$$

Step 5: Loss (Binary Cross-Entropy)

$$\mathcal{L} = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

$$= -1 \cdot \log(0.6900) - 0 \cdot \log(0.3100)$$

$$= -(-0.3711) = \boxed{0.3711}$$

BACKWARD PASS

Step 6: Output Error Signal δ_2

$$\delta_2 = \hat{y} - y = 0.6900 - 1 = \boxed{-0.1100}$$

Step 7: Gradient w.r.t. w_2

$$\frac{\partial \mathcal{L}}{\partial w_2} = \delta_2 \cdot a_1 = -0.1100 \times 0.7503 = \boxed{-0.0825}$$

Step 8: Gradient w.r.t. b_2

$$\frac{\partial \mathcal{L}}{\partial b_2} = \delta_2 = \boxed{-0.1100}$$

Step 9: Sigmoid Derivative at Layer 1

$$a_1(1 - a_1) = 0.7503 \times (1 - 0.7503) = 0.7503 \times 0.2497 = \boxed{0.1873}$$

Step 10: Gradient w.r.t. w_1

$$\frac{\partial \mathcal{L}}{\partial w_1} = \delta_2 \cdot w_2 \cdot a_1(1 - a_1) \cdot x$$

$$= -0.1100 \times 0.8 \times 0.1873 \times 2$$

$$= -0.1100 \times 0.8 = -0.0880$$

$$= -0.0880 \times 0.1873 = -0.01648$$

$$= -0.01648 \times 2 = \boxed{-0.03297}$$

Step 11: Gradient w.r.t. b_1

$$\frac{\partial \mathcal{L}}{\partial b_1} = \delta_2 \cdot w_2 \cdot a_1(1 - a_1)$$

$$= -0.1100 \times 0.8 \times 0.1873 = \boxed{-0.01648}$$

WEIGHT UPDATE

Using $\text{new} = \text{old} - \eta \cdot \text{gradient}$:

Update w_2

$$w_2^{\text{new}} = 0.8 - 0.5 \times (-0.0825) = 0.8 + 0.04125 = \boxed{0.8413}$$

Update b_2

$$b_2^{\text{new}} = 0.2 - 0.5 \times (-0.1100) = 0.2 + 0.0550 = \boxed{0.2550}$$

Update w_1

$$w_1^{\text{new}} = 0.5 - 0.5 \times (-0.03297) = 0.5 + 0.01648 = \boxed{0.5165}$$

Update b_1

$$b_1^{\text{new}} = 0.1 - 0.5 \times (-0.01648) = 0.1 + 0.00824 = \boxed{0.1082}$$

VERIFY: Run Forward Pass Again with New Weights

$$z'_1 = 0.5165 \times 2 + 0.1082 = 1.033 + 0.1082 = 1.1412$$

$$a'_1 = \sigma(1.1412) = \frac{1}{1 + e^{-1.1412}} = \boxed{0.7577}$$

$$z'_2 = 0.8413 \times 0.7577 + 0.2550 = 0.6375 + 0.2550 = 0.8925$$

$$\hat{y}' = \sigma(0.8925) = \frac{1}{1 + e^{-0.8925}} = \boxed{0.7096}$$

$$\mathcal{L}' = -\log(0.7096) = \boxed{0.3430}$$

Complete Backpropagation from Scratch

PART 1: Network Setup

$$z_1 = w_1x + b_1$$

$$a_1 = \sigma(z_1)$$

$$z_2 = w_2a_1 + b_2$$

$$\hat{y} = \sigma(z_2)$$

Loss (Binary Cross-Entropy):

$$\mathcal{L} = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

PART 2: Computation Graph

$$x \rightarrow \boxed{z_1 = w_1 x + b_1} \rightarrow \boxed{a_1 = \sigma(z_1)} \rightarrow \boxed{z_2 = w_2 a_1 + b_2} \rightarrow \boxed{\hat{y} = \sigma(z_2)} \rightarrow \boxed{\mathcal{L}}$$

The graph shows which variables depend on which — this is what drives the chain rule.

Chain Rule for w_2 :

$$\frac{\partial \mathcal{L}}{\partial w_2} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2} \times \frac{\partial z_2}{\partial w_2}$$

Chain Rule for b_2 :

$$\frac{\partial \mathcal{L}}{\partial b_2} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2} \times \frac{\partial z_2}{\partial b_2}$$

Chain Rule for w_1 :

$$\frac{\partial \mathcal{L}}{\partial w_1} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2} \times \frac{\partial z_2}{\partial a_1} \times \frac{\partial a_1}{\partial z_1} \times \frac{\partial z_1}{\partial w_1}$$

Chain Rule for b_1 :

$$\frac{\partial \mathcal{L}}{\partial b_1} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_2} \times \frac{\partial z_2}{\partial a_1} \times \frac{\partial a_1}{\partial z_1} \times \frac{\partial z_1}{\partial b_1}$$

◆ Piece 1: $\partial \mathcal{L} / \partial \hat{y}$

$$\mathcal{L} = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$$

Differentiate term 1:

$$\frac{\partial}{\partial \hat{y}} [-y \log(\hat{y})] = -y \cdot \frac{1}{\hat{y}} = -\frac{y}{\hat{y}}$$

Differentiate term 2 (chain rule: derivative of $1 - \hat{y}$ is -1):

$$\frac{\partial}{\partial \hat{y}} [-(1 - y) \log(1 - \hat{y})] = -(1 - y) \cdot \frac{1}{1 - \hat{y}} \cdot (-1) = +\frac{1 - y}{1 - \hat{y}}$$

Result:

$$\boxed{\frac{\partial \mathcal{L}}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1 - y}{1 - \hat{y}}}$$

◆ Piece 2: $\partial \hat{y} / \partial z_2$ (Sigmoid Derivative)

$$\hat{y} = \sigma(z_2) = \frac{1}{1 + e^{-z_2}} = (1 + e^{-z_2})^{-1}$$

Apply power rule + chain rule:

$$\frac{\partial \hat{y}}{\partial z_2} = -1 \cdot (1 + e^{-z_2})^{-2} \cdot \frac{d}{dz_2}(e^{-z_2})$$

Since $\frac{d}{dz_2}(e^{-z_2}) = -e^{-z_2}$:

$$= -1 \cdot (1 + e^{-z_2})^{-2} \cdot (-e^{-z_2}) = \frac{e^{-z_2}}{(1 + e^{-z_2})^2}$$

Split the fraction:

$$= \frac{1}{1 + e^{-z_2}} \cdot \frac{e^{-z_2}}{1 + e^{-z_2}} = \frac{1}{1 + e^{-z_2}} \cdot \left(1 - \frac{1}{1 + e^{-z_2}}\right)$$

$$\boxed{\frac{\partial \hat{y}}{\partial z_2} = \hat{y}(1 - \hat{y})}$$

◆ Piece 3: $\partial z_2 / \partial w_2$

$$z_2 = w_2 a_1 + b_2$$

$$\boxed{\frac{\partial z_2}{\partial w_2} = a_1}$$

◆ Piece 4: $\partial z_2 / \partial b_2$

$$z_2 = w_2 a_1 + b_2$$

$$\boxed{\frac{\partial z_2}{\partial b_2} = 1}$$

◆ Piece 5: $\partial z_2 / \partial a_1$

$$z_2 = w_2 a_1 + b_2$$

$$\boxed{\frac{\partial z_2}{\partial a_1} = w_2}$$

◆ Piece 6: $\partial a_1 / \partial z_1$ (Sigmoid Derivative again)

$$a_1 = \sigma(z_1)$$

Same derivation as Piece 2, just different variable:

$$\boxed{\frac{\partial a_1}{\partial z_1} = a_1(1 - a_1)}$$

◆ Piece 7: $\partial z_1 / \partial w_1$

$$z_1 = w_1 x + b_1$$

$$\boxed{\frac{\partial z_1}{\partial w_1} = x}$$

◆ Piece 8: $\partial z_1 / \partial b_1$

$$z_1 = w_1 x + b_1$$

$$\boxed{\frac{\partial z_1}{\partial b_1} = 1}$$

PART 5: The Master Simplification (Used Everywhere)

Before plugging into chain rules, simplify Piece 1 $\times$ Piece 2:

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2} = \left(-\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} \right) \cdot \hat{y}(1-\hat{y})$$

Distribute $\hat{y}(1-\hat{y})$:

$$= -\frac{y}{\hat{y}} \cdot \hat{y}(1-\hat{y}) + \frac{(1-y)}{(1-\hat{y})} \cdot \hat{y}(1-\hat{y})$$

Cancel $\hat{y}$ in term 1, cancel $(1-\hat{y})$ in term 2:

$$= -y(1-\hat{y}) + (1-y)\hat{y}$$

Expand:

$$= -y + y\hat{y} + \hat{y} - y\hat{y}$$

Cancel $y\hat{y}$:

$$\boxed{\frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2} = \hat{y} - y = \delta_2}$$

This is called the **output error signal** δ_2 .

PART 6: Plug Into Every Chain Rule

✔ Gradient w.r.t. w_2

$$\frac{\partial \mathcal{L}}{\partial w_2} = \underbrace{\frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2}}_{\delta_2 = \hat{y} - y} \times \underbrace{\frac{\partial z_2}{\partial w_2}}_{a_1}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial w_2} = (\hat{y} - y) \cdot a_1}$$

✔ Gradient w.r.t. b_2

$$\frac{\partial \mathcal{L}}{\partial b_2} = \underbrace{\frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2}}_{\delta_2 = \hat{y} - y} \times \underbrace{\frac{\partial z_2}{\partial b_2}}_1$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial b_2} = (\hat{y} - y)}$$

✔ Gradient w.r.t. w_1

$$\frac{\partial \mathcal{L}}{\partial w_1} = \underbrace{(\hat{y} - y)}_{\delta_2} \times \underbrace{w_2}_{\text{Piece 5}} \times \underbrace{a_1(1 - a_1)}_{\text{Piece 6}} \times \underbrace{x}_{\text{Piece 7}}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial w_1} = (\hat{y} - y) \cdot w_2 \cdot a_1(1 - a_1) \cdot x}$$

✔ Gradient w.r.t. b_1

$$\frac{\partial \mathcal{L}}{\partial b_1} = \underbrace{(\hat{y} - y)}_{\delta_2} \times \underbrace{w_2}_{\text{Piece 5}} \times \underbrace{a_1(1 - a_1)}_{\text{Piece 6}} \times \underbrace{1}_{\text{Piece 8}}$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial b_1} = (\hat{y} - y) \cdot w_2 \cdot a_1(1 - a_1)}$$

PART 7: Weight Update Rules

$$w_2 \leftarrow w_2 - \eta \cdot (\hat{y} - y) \cdot a_1$$

$$b_2 \leftarrow b_2 - \eta \cdot (\hat{y} - y)$$

$$w_1 \leftarrow w_1 - \eta \cdot (\hat{y} - y) \cdot w_2 \cdot a_1(1 - a_1) \cdot x$$

$$b_1 \leftarrow b_1 - \eta \cdot (\hat{y} - y) \cdot w_2 \cdot a_1(1 - a_1)$$

COMPLETE SUMMARY FLOW

1. PREPARE DATA



2. DESIGN ARCHITECTURE



3. INITIALIZE WEIGHTS (random, not zero)



4. FORWARD PASS → get $\hat{y}$



5. COMPUTE LOSS → how wrong?



6. BACKWARD PASS → compute gradients



7. UPDATE WEIGHTS → gradient descent



8. REPEAT steps 4-7 for many epochs



9. EVALUATE on test data



10. TUNE → fix overfitting/underfitting